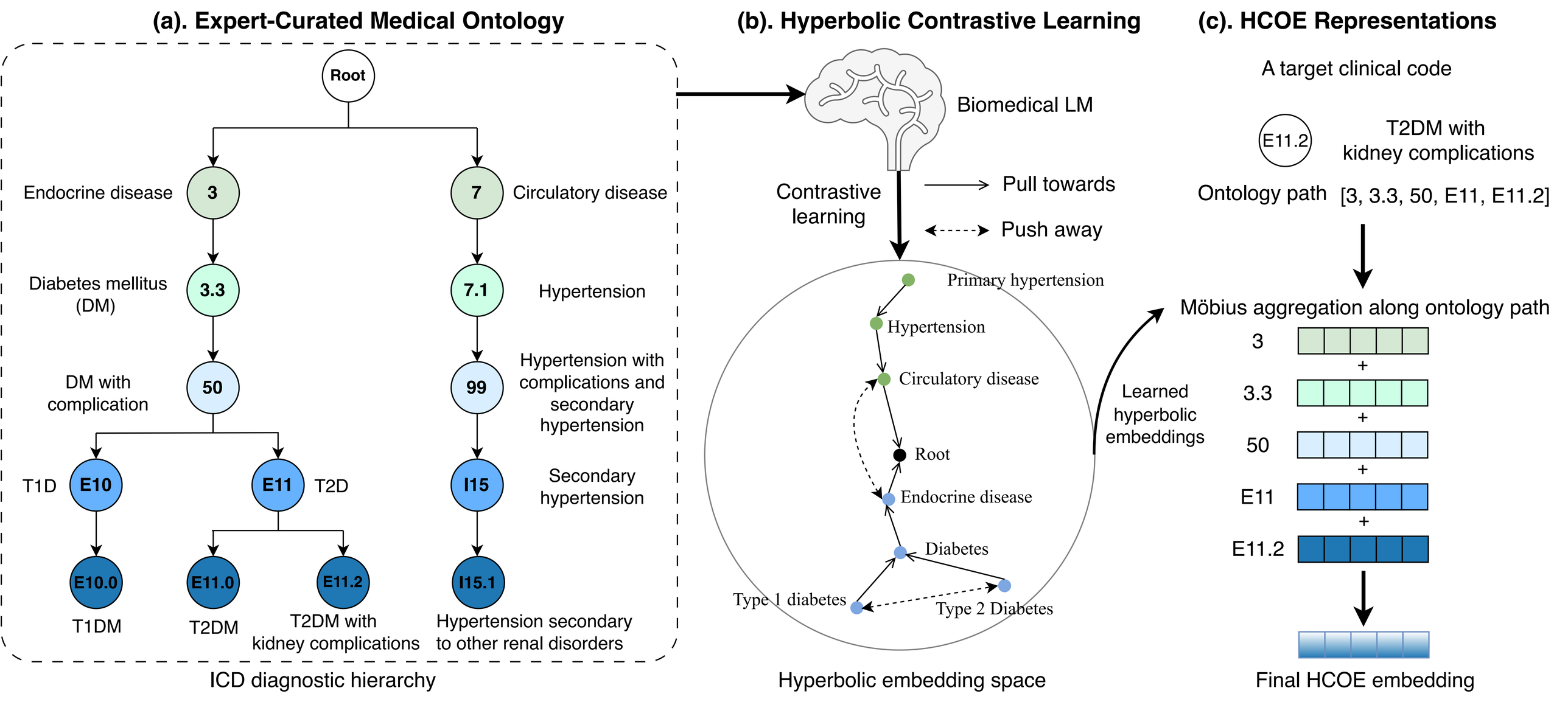

(a). Expert-Curated Medical Ontology
Root
Endocrine disease
3
7
Circulatory disease
Diabetes mellitus (DM)
3.3
7.1
Hypertension
DM with complication
50
99
Hypertension with complications and secondary hypertension
T1D
E10
E11
T2D
I15
Secondary hypertension
E10.0
E11.0
E11.2
I15.1
T1DM
T2DM
T2DM with kidney complications
Hypertension secondary to other renal disorders
ICD diagnostic hierarchy
(b). Hyperbolic Contrastive Learning
Biomedical LM
Contrastive learning
Pull towards
Push away
Primary hypertension
Hypertension
Circulatory disease
Root
Endocrine disease
Diabetes
Type 1 diabetes
Type 2 Diabetes
Hyperbolic embedding space
Learned hyperbolic embeddings
(c). HCOE Representations
A target clinical code
E11.2
T2DM with kidney complications
Ontology path
[3, 3.3, 50, E11, E11.2]
Möbius aggregation along ontology path
3
3.3
50
E11
E11.2
Final HCOE embedding